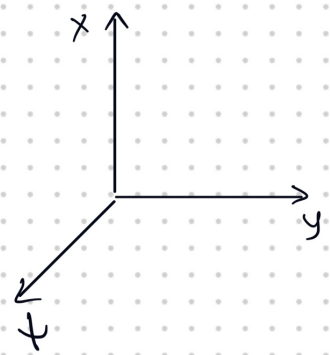
08.09.



$v(t)$

$n(t) = \frac{\omega_n}{g}$

$v(t) = b\sqrt{1 + a^2b^2t^2}$

$n(t) = \frac{ab^2}{g\sqrt{1 + a^2b^2t^2}}$; $a, b = \text{Const}$.

$\dot{x} = v\cos\psi$

$\dot{y} = v\sin\psi$

$\dot{\psi} = v \cdot k$.

Найти: $x(t)$ - ?

$y(t)$ - ?

$\tau = \frac{1}{v} v \begin{pmatrix} \cos\psi \\ \sin\psi \\ 0 \end{pmatrix}$.

$\rho = \frac{1}{k}$

Решение:

$\omega = \dot{v}\bar{\tau} + \frac{v^2}{\rho}\bar{n}$

ω_τ ω_n

$\omega_\tau = 0$

$\omega_n = n(t) \cdot g = g\frac{ab^2}{g\sqrt{1+a^2b^2t^2}} = \frac{ab^2}{\sqrt{1+a^2b^2t^2}} = \omega$

$\omega_n = v^2 k \bar{n}$ единич.

$k(t) = \frac{\omega_n(t)}{v^2\bar{n}} = \frac{n(t)g}{b^2(1+a^2b^2t^2)} = \frac{ab^2 g}{g\sqrt{1+a^2b^2t^2}\, b^2(1+a^2b^2t^2)} =$

$= \frac{a}{(1+a^2b^2t^2)^{3/2}}$.

$\dot{\psi} = v \cdot k = b\sqrt{1+a^2b^2t^2} \cdot \frac{a}{(1+a^2b^2t^2)\sqrt{1+a^2b^2t^2}} = \frac{ab}{1+a^2b^2t^2}$.

$\psi = \int \dot{\psi}\, dt = \int \frac{ab\, dt}{1+a^2b^2t^2} = ab\int \frac{dt}{1+a^2b^2t^2} = ab\int \frac{dt}{\frac{1}{a^2b^2} + t^2} =$

$= \frac{ab}{ab} \operatorname{arctg}\frac{t}{\frac{1}{ab}} + C = \operatorname{arctg} abt + C$

$x = \int b\sqrt{1+a^2b^2t^2}\cos(\operatorname{arctg} abt)\, dt = b\int \sqrt{1+a^2b^2t^2}\frac{dt}{\sqrt{1+a^2b^2t^2}}$

$\cos(\operatorname{arctg} x) = \frac{1}{\sqrt{1+x^2}}$

$\sin(\operatorname{arctg} x) = \frac{x}{\sqrt{1+x^2}}$

$= b \int dt = bt$

$y = \int v \sin\psi = \int b\sqrt{1+a^2b^2t^2} \cdot \frac{abt}{\sqrt{1+a^2b^2t^2}} dt = ab^2 \int t\,dt =$

$= ab^2 \frac{t^2}{2}$

$x(t) = bt$

$y(t) = \frac{ab^2t^2}{2}$